

TWO INTEGRAL DEGENERATION TYPES OF THE SIGNED ERDŐS–STRAUS QUARTIC

THE CLANKERS

ABSTRACT. We specialize the finite signed completion of the literal-root ES quartic at its original prime. Its integral normalization index is p^2 for Type I and p^9 for Type II, despite constant rank eight. The original coordinate conductors, Smith factors and generic rational-point counts are explicit. The new Type-I boundary point lifts modulo p^2 but never modulo p^3 . These statements compare original arithmetic states; they do not prove an ES state exists at every prime.

The original integral signed order and its exact normalization

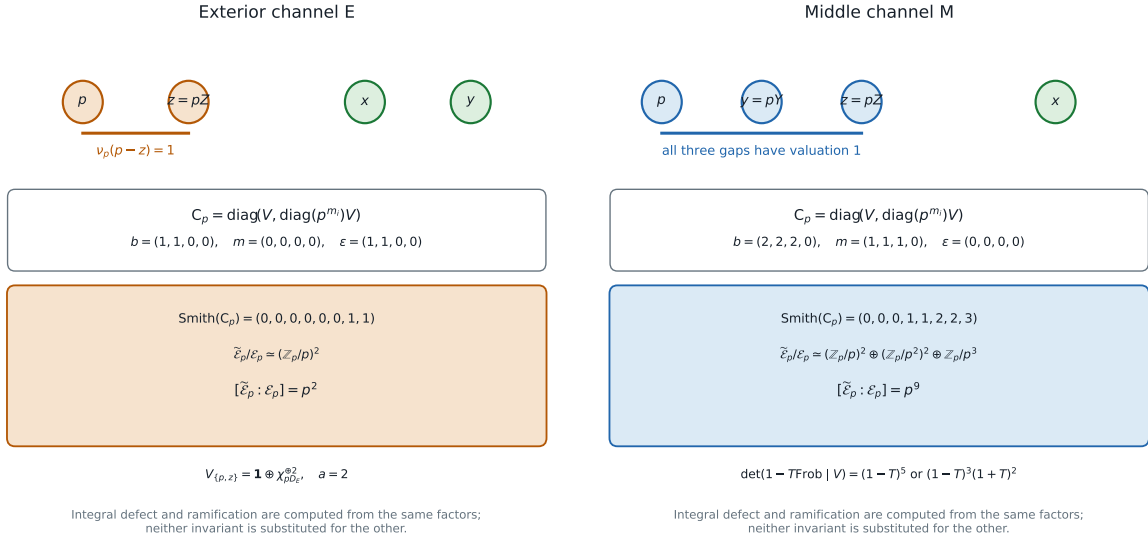


FIGURE 1. The exact normalization matrix in the original ordered basis. Its Smith exponents give index p^2 in Type I and p^9 in Type II; the complete matrix proof and conductor ideals follow below.

1. THE ORIGINAL SOURCE AND ITS FINITE SIGNED COMPLETION

Let $p \equiv 1 \pmod{12}$ be prime. Retain $p/4 < a < p/2$, $R = 4a - p$, $u \mid a^2$, and either $E : R \mid 4u + 1$ or $M : R \mid u + a$. With

$$d = \gcd(a, u), \quad (h, r, s) = (d^2/u, u/d, a/d),$$

the original Type-I/II denominators are

$$E : (a, h\kappa, phr\kappa), \quad \kappa = (pr + s)/R; \quad M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R.$$

These inherited coordinates [1, §2] retain $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. The ordered inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$. Reciprocity gives $(l/R) = (l/p)$ for every prime $l \mid a$, including two by the supplementary law. Hence

$$(1) \quad E : (h/p) = -1, \quad M : (rs/p) = -1.$$

The M claim uses $r \equiv -s \pmod{R}$.

Set $\mathfrak{o} = \mathbb{Z}_p$, $K = \mathbb{Q}_p$, and retain the literal ordered roots

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (p, x, y, z), \quad S = p + x + y + z, \quad H(T) = -S^{-1} \prod_i (T - \ell_i).$$

The cubic coefficient of H is one. The given signed-completion construction [2, §3, (18)–(24)] is

$$(2) \quad \widehat{S} = \mathfrak{o}[T, \eta]/(H, \eta^2 - H'),$$

with original basis $1, T, T^2, T^3, \eta, \eta T, \eta T^2, \eta T^3$. It recovers the original affine inverse only on $\eta \neq 0$, with $\xi = \eta^{-1}$. We prove below that S is a p -unit, so (2) is indeed finite free of rank eight over this original arithmetic ring.

For E write $(x, y, z) = (a, b, pc)$. The original identity $pD + 1 = 4hr\kappa$ gives $c \equiv 1/4 \pmod{p}$. Both $a - b = hs(r - \kappa)$ and $a + b = hs(r + \kappa)$ are units: the contrary would make the nonresidue hr^2 equal to $1/4$ or $-1/4$ modulo p . Thus only ℓ_0, ℓ_3 have a nonunit difference, of valuation one, and $S \equiv a + b$.

For M write $(x, y, z) = (a, pb, pc)$. Here b, c are units, $4bc \equiv b + c$, and $(bc/p) = -1$ by (1). They are distinct modulo p because $r, s < p$ are distinct. Neither is one: then the other would be $1/3$, contrary to $(3/p) = 1$. Thus the cluster ℓ_0, ℓ_2, ℓ_3 has all pair differences of valuation one, its singleton a is separated by units, and $S \equiv a$.

Writing $d_i = H'(\ell_i)$, we have therefore proved

$$(3) \quad E : (v_p(d_i)) = (1, 0, 0, 1), \quad M : (v_p(d_i)) = (2, 0, 2, 2).$$

2. COMPLETE NORMALIZATION AND CONDUCTOR

Let $k_i = \lfloor v_p(d_i)/2 \rfloor$ and

$$N_i = \mathfrak{o}[\theta_i]/(\theta_i^2 - d_i/p^{2k_i}), \quad N = \prod_i N_i.$$

Each N_i is maximal: its quadratic is either a split unit quadratic, an unramified unit quadratic, or an Eisenstein quadratic of odd residue characteristic. A direct proof is obtained by taking trace and norm of $a + b\theta_i$: trace forces $a \in \mathfrak{o}$, then norm forces $b \in \mathfrak{o}$. In the split case use the two values and the unit $2\sqrt{d_i/p^{2k_i}}$. The exact normalization map is

$$(4) \quad f(T) + \eta g(T) \mapsto (f(\ell_i) + p^{k_i} g(\ell_i) \theta_i)_i.$$

Theorem 2.1. *For Type I,*

$$N/\widehat{S} \simeq (\mathfrak{o}/p)^2, \quad \mathfrak{f} = pN_0 \times N_1 \times N_2 \times pN_3.$$

For Type II,

$$N/\widehat{S} \simeq (\mathfrak{o}/p)^2 \oplus (\mathfrak{o}/p^2)^2 \oplus \mathfrak{o}/p^3, \quad \mathfrak{f} = p^3 N_0 \times N_1 \times p^3 N_2 \times p^3 N_3.$$

Thus the normalization indices are p^2 and p^9 , respectively.

Proof. Unit resultants split off the noncolliding root factors integrally. For E the remaining root lattice is $L_2 = \{(v_0, v_3) : v_0 \equiv v_3 \pmod{p}\}$, by linear interpolation across a root difference of exact valuation one. Its Smith factors are $1, p$. Equation (4) gives $L_2 \oplus L_2$, and four already maximal outside coordinates. A normal value supported at one cluster root belongs to the order precisely when both its coefficients in $1, \theta_i$ are divisible by p . This gives the quotient and maximal conductor ideal in the Type-I case.

For M use the Newton basis $1, T - p, (T - p)(T - pb)$ on the three clustered roots. Evaluation has Smith factors $1, p, p^2$, so its root lattice L_3 has index p^3 . Equation (4) gives $L_3 \oplus pL_3$, with exponents $(0, 1, 2, 1, 2, 3)$, and two outside zeros. A value $a + b\theta_i$ supported at one cluster root has Lagrange denominators of order p^2 . Thus membership requires $a \in p^2\mathfrak{o}, b \in p^3\mathfrak{o}$. Membership after multiplication by the unit θ_i forces both into $p^3\mathfrak{o}$. That condition is also sufficient, proving the Type-II conductor and quotient. $\square$

In particular the eight Smith exponents are

$$(0, 0, 0, 0, 0, 0, 1, 1) \quad \text{and} \quad (0, 0, 0, 1, 1, 2, 2, 3).$$

The full inverse of (4) interpolates the values a_i and b_i/p^{k_i} separately. It is integral exactly when all eight original polynomial coefficients are integral. The nonzero finite quotient is retained; no $\mathfrak{o}$ -linear section from that torsion quotient into the torsion-free N exists. These conductors are conductors of finite orders, not the analytic weighted conductor used elsewhere in the programme.

3. GENERIC SIGNED POINTS AND SPECIAL BOUNDARY POINTS

Over K , (2) is $\prod_i K[\eta_i]/(\eta_i^2 - d_i)$. For E the two valuation-one square classes are equal, because their unit ratio is $-1 \pmod{p}$. At the other two roots the derivative residues are $-a^2(a-b)/(a+b)$ and $-b^2(b-a)/(a+b)$, also with square ratio. Thus the fibre has either four K -points or none; its nonsplit factors are two copies of a ramified quadratic and possibly two copies of an unramified quadratic. For M, the singleton derivative is $-a^2 \pmod{p}$. The three cluster derivatives after division by p^2 have residues

$$(1-b)(1-c), \quad (b-1)(b-c), \quad (c-1)(c-b).$$

Their product is a negative square. Either all three or exactly one is square, so the complete fibre has eight or four K -points. Unit square roots lift by $z_{n+1} = z_n - p^n((z_n^2 - d)/p^n)(2z_n)^{-1} \pmod{p^{n+1}}$. This accounts for every signed point and every field factor.

The colliding special blocks are respectively

$$\mathbb{F}_p[\eta]/\eta^4, \quad \mathbb{F}_p[\epsilon, \eta]/(\epsilon^3, \eta^2 - 3\epsilon^2).$$

Their lengths are four and six, while multiplication by η has Jordan blocks (4) and (4, 2). The original inverse obtained by inverting η deletes these entire nonzero local blocks; it does not make their poles finite.

Theorem 3.1. *The added Type-I point $(T, \eta) = (0, 0) \pmod{p}$ has exactly p lifts modulo p^2 , but no lift modulo p^3 .*

Proof. Factor $H(T) = U(T)(T-p)(T-pc)$, where $U(0) = -ab/S$ is a unit and $c = 1/4 \pmod{p}$. For $T = pw, \eta = pv$, the root equation is automatic modulo p^2 , and the quadratic equation requires $2w - 1 - c = 0 \pmod{p}$. Thus $w = 5/8$ and all p choices of v work. But every next lift has

$$H(T)/p^2 \equiv -U(0)(1-c)^2/4 = 9ab/(64S) \not\equiv 0 \pmod{p}.$$

The root equation fails already modulo p^3 . □

At the hard prime 1009, the original E denominators (253, 87032, 3818056) have *no* K -rational signed inverse, while the original M denominators (253, 2042216, 88792) have eight. Both are positive integer ES identities. Thus degree-eight finite flatness is not rational-point or integral-section surjectivity. The proper map retains length at its boundary; it does not discharge the prime-by-prime original occupancy problem.

The distinguished-prime idempotent

$$e_p(T) = \frac{\prod_{i=1}^3 (T - \ell_i)}{\prod_{i=1}^3 (p - \ell_i)}$$

has an exact denominator p in E and p^2 in M. Hence the generic prime-only sign map $\eta \mapsto (1 - 2e_p)\eta$ does not preserve the original integral order, whereas the global sign does. Both extend to the normalization. This states the actual loss instead of deleting the original prime-root marking.

4. PROVENANCE AND EXECUTION BOUNDARY

The literal target, finite rank-eight extension and open inverse are supplied antecedents [2]. The indices, original conductors, local signed types, and sharp finite lifting obstruction are derived here. No historical-priority claim is made. The full companion proof gives all coordinate maps, the reciprocal quadratic twist, and explicit finite-place rational controls satisfying the supplied height inequalities but missing one global denominator condition. It also transports the previously proved $p = 825241$ capacity crossing; that crossing is supplied by original divisor arithmetic, not by assuming an inverse point of this finite cover.

The package's standard-library verifier and a separate primitive-source checker compare every labelled E/M state at all primes $p \equiv 1 \pmod{12}$ through 3000. The detailed original lattice and inverse examples, finite special fibres, negative controls and command receipts are included. Finite replay does not prove initial ES occupancy, a Lean build or an independent mathematical review.

REFERENCES

- [1] C. Elsholtz and T. Tao (2013), *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. 94, 50–105, §2. <https://arxiv.org/abs/1107.1010>.
- [2] The Clankers / Split-Zero continuation (2026), supplied `Pasted markdown(20260920-025519).md`, §§1–3, especially equations (8)–(11) and (18)–(24). Original source preserved with this package.